INFRASTRUCTURE SUPPORT PROCESS

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CHANGE HISTORY

This section describes the details of changes that have resulted in the current document.

SL. No.	Document Version No.	Change Description	Reason For change	Date Prepared By	Date Reviewed By	Date Approved By
1	0.1	Initial Draft		Anitha N 21-Apr-25		
2	0.2				Santosh 08-May-25	
3	Issue B, V1.0					Sanmangala M S 13-Jun-25
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1 Purpose

This document outlines the processes and responsibilities associated with Level 1 (L1) and Level 2 (L2) support for managing infrastructure to ensure efficient resolution of issues, minimal downtime, and effective communication.

2 Scope

This process applies to all L1 and L2 support for cloud-hosted infrastructure services, infrastructure components, and user interactions across environments.

3 Abbreviations

a. KEDB -Known error Database

b. RCA – Root Cause Analysis

4 Terms and Definitions

a. L1 Support: First-line support team responsible for monitoring, basic issue resolution, and escalation.

b. L2 Support: Second-line support team responsible for advanced troubleshooting, configuration changes, and root cause analysis.

c. L3 Support: Third-line support team responsible for development and bug fixes

d. Incident: Any event which is not part of the standard operation and causes, or may cause, an interruption or degradation of service.

5 Responsibilities

a. L1 Support Team

- Monitor dashboards and alerts from monitoring tools.
- Perform initial triage and basic troubleshooting.
- Resolve common issues (e.g., service restarts, access issues).
- Log all incidents with detailed information.
- Escalate unresolved or complex issues to L2.

b. L2 Support Team

- Investigate escalated incidents.
- Perform advanced diagnostics and system-level troubleshooting.
- Implement cloud configuration changes (e.g., network policies, security groups).
- Coordinate with L3 teams or vendors as needed.

- Document RCA and update SOPs.

c. Lead

- Lead, supervise, and support the technical support team during the shift.
- Ensure the team is meeting performance goals, including response time, resolution time, and customer satisfaction.
- Monitor incoming support tickets, calls, or chats to ensure issues are being addressed in a timely manner.
- Provide ongoing coaching and training to the support team, ensuring they have up-to-date knowledge on technical products, tools, and processes.
- Ensure all technical issues, solutions, and customer interactions are accurately documented in the ticketing system.

d. Project Manager

- Develop detailed project plans that outline the scope, goals, timelines, resources, and budget required.
- Set realistic timelines and ensure that all team members understand their responsibilities and deadlines.
- Manage client relationships throughout the project lifecycle, ensuring customer satisfaction and addressing concerns promptly.
- Manage the project budget, ensuring costs are kept within approved limits while ensuring the necessary resources are available.

6 RACI

Process	Responsible	Accountable	Consulted	Informed
Incident Detection and Logging	Monitoring Tool, Helpdesk System	L1 Support	Lead	Project Manager
Initial investigation and Basic Troubleshooting	L1 Support	Lead	L2 Support	Project Manager
Escalation to L2	L1 Support	Lead	--	L2 Support, Project Manager
Advanced Troubleshooting	L2 Support	Lead	L3 Support	Project Manager

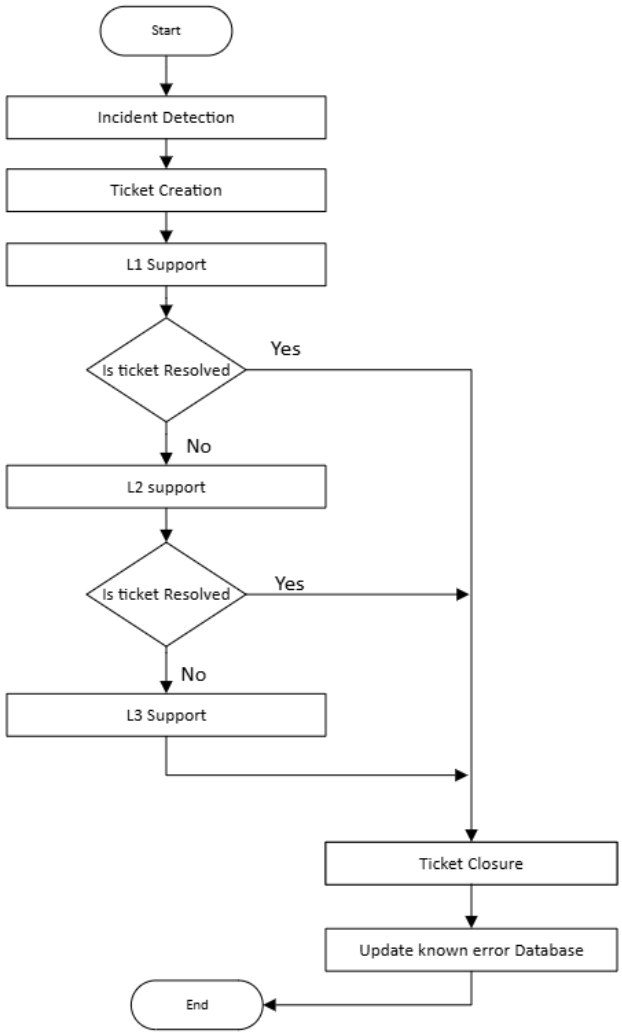
Escalation to L3/Vendor	L2 Support	Lead	Project Manager	L1, L3 Support, Project Manager
Known Error Analysis and record creation	L1, L2 Support	Knowledge Manager	L1 & L2 Support, Lead	Project Manager

7 SIPOC

Supplier	Input	Process and Control		Output	Customer
		Process	Control		
End Users, Monitoring tool	Incident report/service request	Incident Detection and Logging	Monitoring tool alerts, ticketing rules	Ticket created	L1 Support
L1 Support	Ticket details, access to knowledge base	Basic trouble shooting, Ticket Tracking and Status Management	SLA, ticket workflow, known error DB, troubleshooting checklist	Updated ticket status, Fix applied, or escalation initiated	Project Manager, Lead
L1 Support	Unresolved issue, Escalated ticket, logs, diagnostics	Escalation to L2, Advanced Troubleshooting	Escalation matrix	Fix/work around or escalate to L2	Lead, L2 Support
L2 Support	Issue details, system access	Advanced Analysis and Fix (L2)	SOPs, Infra Team policies	Resolved issue or escalation	End Users, Lead
L2 Support	Unresolved issue	Escalation to L3/Vendor	Escalation path, vendor SLAs	Escalated case	L3 Support, Vendors
L1, L2 Support	Repeated incident, logs	Known Error Analysis and Record Creation	KEDB maintenance policies	KEDB entry created	Knowledge Manager, Lead

8 Overall Process Flow Chart

a) Infrastructure Support Process



9 Process Description

9.1 Incident Detection

a. Alert generated via monitoring tools (AWS CloudWatch, Azure Monitor, Datadog) which automatically detects and converts into a ticket or a user raises a ticket in the ticketing tool (e.g Jira, Service now).

b. The ticketing tool will have below mentioned details:

- Subject: A concise description of the issue or request.
- Requester Information: Name, email address, and potentially other contact details of the person submitting the ticket.
- Description: A more detailed explanation of the issue or request.

- Priority: A level of urgency assigned to the ticket, based on the impact of the issue.
- Assignment: The helpdesk system may automatically assign the ticket to an appropriate agent or team based on predefined rules.

c. Ticket Tracking and Management:

- **Status Updates:**
  - The Tool tracks the status of the ticket, allowing agents to update it with progress and notes.
- **Prioritization:** Tickets can be prioritized based on various factors, such as impact or urgency. severity levels categorize the impact of a ticket, guiding priority and resolution efforts. Typically, these levels range from critical (highest impact) to low (minimal impact), with common examples like "Critical," "Major," "Minor,". These levels help determine response times and escalate issues appropriately. Common Severity Levels:
  - Critical: This indicates a major issue with a significant business impact, potentially halting operations and requiring immediate attention. Examples include a system outage or critical data loss.
  - Major: A major issue that impacts a large portion of the business, causing significant performance degradation or functionality loss. While operations may continue in a limited way, a workaround may not be readily available.
  - Minor: An issue that affects a smaller portion of the business or has a minor performance impact, but a workaround may be available.

- Severity names can change based on the customer requirement.
- **Communication:**
  - Agents can communicate with users through the ticket system, requesting more information or providing updates.
- **Resolution:**
  - Once the issue is resolved, the ticket is marked as resolved and potentially closed.

d. When an issue is escalated to a higher level, the severity level often dictates which specialists or teams are involved.

9.2 L1 Support Handling

a. Ticket assigned to L1 team.

b. L1 performs initial investigation by referring to their respective guidelines.

- Check logs
- Check Known error database
- Restart service if needed
- Validate access/configuration

c. If resolved:

- Update the ticketing tool and close ticket

d. If unresolved:

- Document all findings in the tool
- Escalate to L2 with ticket notes and logs

9.3 L2 Support Handling

a. Review escalated ticket details.

- Conduct advanced analysis as per the plan
- CLI/Cloud console investigation
- System log review
- Network checks

b. Apply fix or workaround.

c. Capture, root cause and resolution.

d. Update or create knowledge base article if applicable.

e. Update the Known Error database, for new errors and solutions

f. Close the ticket or escalate to L3 if required.

9.4 L3 Support handling

a. L3 support may involve code or development related fixes and require collaboration with developers or vendors to find comprehensive solutions (not in scope of this SOP, refer SOP-Life Cycle Services Process)

9.5 Known Error database

Known Error Database (KEDB) stores documented information about known errors, including their root cause, workarounds, and related incidents

a. Incident Analysis: When a ticket is logged, it is analyzed to determine if it is caused by a known error.

b. Record Creation: If a known error is identified, a record is created to document the error, its root cause, and any available workarounds.

c. Knowledge Article Creation: A knowledge article is often created from the issue record, making the information easily accessible to agent.

d. Workaround Implementation: The identified workaround is communicated to relevant users.

e. Ticket Resolution: Once the workaround is applied, the ticket is resolved, and the issue record is updated with the status of the known error.

f. Ongoing Maintenance: The KEDB is continuously updated with new information about known errors and their resolutions.

9.6 Escalation Matrix

Tickets can be escalated through multiple tiers, typically starting with Level 1 support (frontline) and progressing to Level 2 (technical support), Level 3 (Development team). Detailed escalation matrix is available in guidelines.

9.7 Tools & Platforms

Purpose	Tool Examples
Monitoring	AWS CloudWatch, Azure Monitor, Datadog
Logging	CloudTrail, ELK, Splunk
Ticketing	ServiceNow, Jira
Access Management	IAM, Azure AD, Okta
Communication	Slack, Teams, Email

10 Information Security requirements

a. The information security related requirements shall be included in the requirements for new information systems or enhancements to existing information systems.

b. Information involved in application services passing over public networks shall be protected from fraudulent activity, contract dispute and unauthorized disclosure and modification.

c. Information involved in application service transactions shall be protected to prevent incomplete transmission, misrouting, unauthorized message alteration, unauthorized disclosure, unauthorized message duplication or replay.

d. Rules for the development of software and systems shall be established and applied to developments within the organization.

e. Changes to systems within the development lifecycle shall be controlled using formal change control procedures.

f. When operating platforms are changed, business critical applications shall be reviewed and tested to ensure there is no adverse impact on organizational operations or security.

g. Modifications to software packages shall be discouraged, limited to necessary changes and all changes shall be strictly controlled.

h. Principles for engineering secure systems shall be established, documented, maintained, and applied to any information system implementation efforts.

i. Organizations shall establish and appropriately protect secure development environments for system development and integration efforts that cover the entire system development lifecycle.

j. The organization shall supervise and monitor the activity of outsourced system development.

k. Testing of security functionality shall be carried out during development.

l. Acceptance testing programs and related criteria shall be established for new information systems, upgrades, and new versions.

m. Test data shall be selected carefully, protected, and controlled.

n. Confidentiality of data is maintained.

11 Metrics

a. Number of incidents handled by L1 and L2

b. Average resolution time

c. Escalation rate from L1 to L2

d. SLA adherence

12 References

SOP-Life Cycle Services Process UTSDLCSSOP

13 Annexures

NA

14 Forms

1. SLA Tracking Template UTSDISPTMP01